25900353 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCECT200, MG3DSCECC200 - ANALOG ELECTRONICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. In a voltage divider network collector current Ic is mainly determined by [K] / [CO1]

   a). Vcc and Rc b). R1 and R2

   c). Re d). None of these

2. The barkhausen criterion requires the feedback signal to be in [K] / [CO1]

   a). Phase with the input b). Out of phase with the input

   c). Same amplitude with the input d). All the above

3. In an RC differentiator ,the output voltage is proportional to [U] / [CO1]

   a). Integral of the input voltage b). The square of input voltage

   c). The derivative of input voltage d). The input voltage itself

4. In a BJT, which region is heavily doped? [U] / [CO1]

   a). Base b). Emitter

   c). Collector d). Emitter and Collector

5. The minimum closed-loop gain of a non-inverting amplifier is [U] / [CO2]

a). 0 b). 0.5 c). 1 d). Infinite

6. The input offset voltage is.. [K] / [CO2]

a). The voltage at the output when both the inputs are zero

b). The voltage needed between inputs to make output zero

c). The differnce in power supply voltages

d). Always zero

7. What is the function of PIN 7 in a 741 op amp [K] / [CO2]

a). Output

b). Inverting input

c). V+(Positive power supply)

d). ground

8. The loop gain requirement for sustaining oscillations in an RC phase shift oscillator is [An] / [CO3]

a). $A \geq 1$

b). $A \geq 10$

c). $A \geq 29$

d). $A \geq 3$

9. In a non-inverting Schmitt trigger, the input signal is applied to [K] / [CO3]

a). Inverting terminal

b). Non Inverting terminal

c). Both terminals

d). None of the above

10. Which feedback is used in an op-amp comparator? [U] / [CO3]

a). Negative feedback

b). Positive feedback

c). No feedback

d). Both positive and negative

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Why RC coupling is preferred in multi stage amplifier design? [U] / [CO1]

12. Briefly explain the Saturation mode of operation of a BJT [U] / [CO1]

13. Compare inverting and non-inverting amplifiers in terms of gain, phase shift [U] / [CO2, CO3]

14. Draw the block diagram of an Op-amp. [U] / [CO2]

15. Why is the open-loop op-amp gain considered impractically high for analog signal amplification? [U] / [CO2, CO3]

16. Explain how a differentiator circuit can be consructed using op amp? Give its output waveforms too. [U] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Explain the working principle of an RC phase shift oscillator using a BJT [U] / [CO1]

18. What is the operating point of a BJT? Explain.Why is it so important? [U] / [CO1]

19. Compare the transition from SSI to VLSI in integrated Circuit? [K] / [CO2]

20. Describe the astable mode operation of the 555 timer IC with a circuit example. Write the equations for frequency and duty cycle. [U] / [CO3]

[10x2 = 20]